Cambridge
O Level

Cambridge International Examinations
Cambridge Ordinary Level

ENVIRONMENTAL MANAGEMENT **5014/22**

Paper 2 **May/June 2017**

MARK SCHEME

Maximum Mark: 60

Published

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Question	Answer	Marks
1(a)(i)	(900 / 25.7 =) 35;	**1**
1(a)(ii)	25;; (*if answer incorrect, allow one mark for 35 × 2 or = 70 [1]*)	**2**
1(a)(iii)	*any one of:* saves buying them; to make more profit; AVP, e.g. recycling / environmental issue / small bags are scarce / expensive to buy;	**1**
1(a)(iv)	*any two of:* sand stolen; sand needs to be guarded at night; danger to life; AVP;	**2**
1(b)	*any two of:* vegetables supply; vitamins or any named vitamin; minerals or any named mineral; roughage / fibre; essential calories / supplies carbohydrate or protein or fat; prevent malnutrition; balanced diet;	**2**

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Question	Answer	Marks
2(a)(i)	(**A**) anemometer; (**B**) rain gauge;	**2**
2(a)(ii)	thermometer / sunshine recorder / barometer / wind vane;	**1**
2(a)(iii)	*any two of:* grass surface; flat / stable ground; clear ground / away from trees or buildings / eq;	**2**
2(a)(iv)	7;	**1**
2(a)(v)	orientation and linear scale; *x*-axis fully labelled; *y*-axis fully labelled; all plots correct;	**4**
2(a)(vi)	April OR May OR June to September OR October;	**1**
2(a)(vii)	*any two of:* cheap or easy to construct; does not use too much labour / can be built by local people or farmers; more water bodies to feed nearby fields / eq; less relocation of people; *idea of* less flooding; AVP, e.g. easier to irrigate crops / greater access to water (to grow crops) / more sustainable / easier to repair / less pollutants likely to accumulate;;	**2**
2(a)(viii)	*any one of:* warm enough for plant growth; only a short dry season / eq;	**1**

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Question	Answer	Marks
2(b)(i)	(solar / heat) energy absorbed at the surface / eq;	**1**
2(b)(ii)	0.0–0.5 m / 0–50 cm;	**1**
2(b)(iii)	(reversed shape due to) radiative or longwave cooling / cools most from the surface / it's the surface where the longwave radiation comes from / ORA / eq;	**1**
2(b)(iv)	*any three of:* *reference to* photosynthesis; further detail related to photosynthesis, e.g. optimal temperature for photosynthesis / photosynthetic rate increases / eq; *reference to* enzymes; further detail related to enzymes, e.g. discusses optimal temperature for enzymes; faster growth possible during day or when warm; some growth at night or cell elongation;	**3**
2(c)(i)	*after first year of cultivation* (rapid) drop from centre to edge of garden or from **P** to **T**; *after three years of cultivation* same availability across garden / eq;	**2**
2(c)(ii)	*any two of*: more people / population increases; more food needed; no more land (can be cultivated); AVP;	**2**
2(c)(iii)	*any two of:* not enough or loss of, nutrients / nitrates or other named minerals / eq; *reference to* damaging soil structure / soil degradation; poor plant growth; plant prone to disease;	**2**

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Question	Answer	Marks
2(d)(i)	table drawn; suitable headings; minimum of 4 cells to collect data;	**3**
2(d)(ii)	*any three of:* not all paths surveyed; only for, 3 hours / limited time; only, some / two days a week; do not know quantity or mass or weight of wood / only people counted; AVP;	**3**
2(d)(iii)	*any three of:* use more people to do the survey / eq; survey all or more paths; more days; more hours / more times, a day; weigh wood / other measurement; AVP;	**3**
2(e)	*any two of:* being burnt (by the fire); being scalding (by the contents); damage to lungs / respiratory or breathing difficulties / named lung condition / eq; damage to eyes; toxic or poisonous gases / carbon monoxide; food poisoning / undercooked food / named illness;	**2**

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>2(f)(i)</td><td>6 correct [3]
4 to 5 correct [2]
2 to 3 correct [1]

<table><tr><th>open fire</th><th>new type of stove</th></tr><tr><td>5.75 (allow 5.5 to 6.0)</td><td>3.75 (allow 3.5 to 4.0)</td></tr><tr><td>15</td><td>10</td></tr><tr><td>5</td><td>13</td></tr></table>;;;</td><td>3</td></tr>
<tr><td>2(f)(ii)</td><td>256;</td><td>1</td></tr>
<tr><td>2(f)(iii)</td><td>any two of:
same volume of water;
same quantity or mass of hay;
same quantity or mass of beans;
same size of box;
same size pot;
type or species of wood;
AVP;</td><td>2</td></tr>
<tr><td>2(f)(iv)</td><td>14 / 14.1 / 14.09;;

(if answer incorrect, allow one mark for correct working or 42, e.g. 298-256 = 42 [1])</td><td>2</td></tr>
<tr><td>2(g)(i)</td><td>any three of:
less wood used / wood saved;
fewer trees cut down;
trees can be replanted;
wood is a renewable resource / eq;
more food can be cooked (to supply an increase in population);
less carbon dioxide emissions / reduced impact on climate / it’s carbon neutral;
save time for cooking or gathering wood / cooks faster;
AVP, e.g. appropriate (sustainable) technology;</td><td>3</td></tr>
</table>

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Question	Answer	Marks
2(g)(ii)	*any two of:* (limited) supply **and** (high) demand idea / ORA; (more) transport costs; AVP;	**2**
2(g)(iii)	*any two of:* not possible to cook on electricity due to cost; power cuts; supply of other fuel is unreliable; they know how to do it; reduce (running) costs / other fuel is more expensive; cools more slowly / keeps food hot for longer / cooks faster; safe to use;	**2**

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